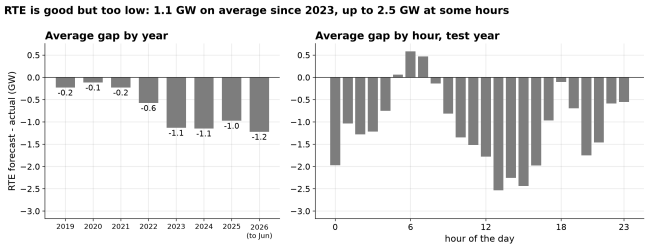
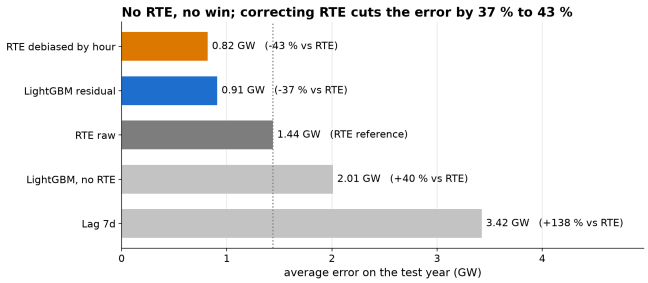
Day-ahead load forecast: correct RTE, don't replace it

French national consumption · 24 hours of D+1 sent on D at 12:00 · test year Jul 2025 to Jun 2026

Bottom line. Our models never beat RTE's public forecast on their own (best: 40 % worse). But RTE has been too low by about 1.1 GW since 2023, by an amount that depends on the hour. Correcting it cuts the error by 37 % (LightGBM on RTE's residual, final model) to 43 % (hourly debiasing rule).

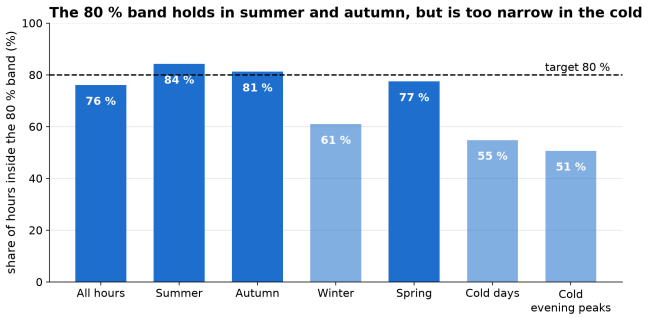
Forecast	All hours	Evening peak	Cold evening peaks	vs RTE raw
RTE raw	1,439 MW	1,348 MW	2,307 MW	reference
LightGBM residual	910 MW	987 MW	1,906 MW	−36.8 %
RTE debiased by hour	819 MW	943 MW	1,528 MW	−43.1 %

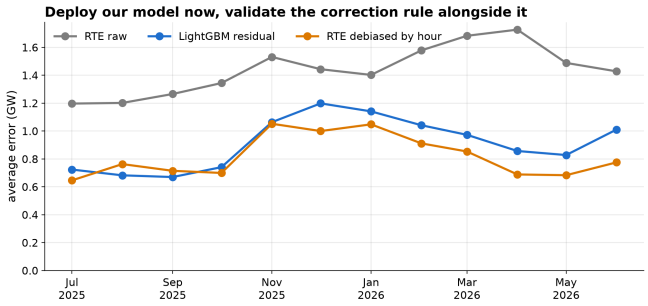




1. Can we beat RTE, and by how much?

Yes, by correcting it. Final model: 910 MW of MAE vs 1,439 MW for RTE (-36.8 %), and better on every scope, including cold evening peaks (1,906 MW vs 2,307 MW). The rule "RTE minus its mean error at the same hour over the last 28 known days" does even better: 819 MW. Same ranking on the walk-forward folds used for every choice (830 MW, 860 MW, 1,362 MW).





2. How uncertain, and when does it fail?

The 80 % band is about 2.4 GW wide and holds 76 % of test hours, but only 61 % in winter and 51 % on cold evening peaks: **it must be wider on cold days.** The model fails when RTE fails (Christmas week, holidays, cold days, some weekends): it removes RTE's usual error, not its error of the day.

3. What to deploy, and the risks

Deploy the LightGBM residual now (selected before the test, with uncertainty bands) and **run the hourly debiasing rule alongside it** as a candidate (simpler and better, but no bands yet).

- **RTE published after noon** (assumed, not verified): check RTE's calendar, fall back to our model without RTE.
- **RTE's bias shifts**: monitor the 7-day bias, alert beyond ±0.5 GW, retrain regularly.
- **Live vs consolidated data**: replay on archived real-time data before go-live.
- **Extreme cold, unusual holidays**: manual override and a wider safety margin.